ACADEMIC HACKATHON EDITION 2026 • CHAMBER N° 001

Make a *physical connection.*

IOTogether connects schools, colleges, laboratories, and makerspaces possessing idle electronics equipment with verified student innovators and engineering teams. Powered by single-sided 3D rope physics, physical ID inspection, and dual-screen QR handover tokens.

3	37	61	15 / 15	100%
PARTNER INSTITUTIONS	APP ROUTER ROUTES	PBD VERLET NODES	AUTOMATED TESTS PASSED	INTEGER PAISE PRICING

1. Problem Statement & Architectural Mission

01 / OVERVIEW

The Electronics Resource Paradox

Engineering colleges and school robotics labs routinely possess high-grade electronics—digital storage oscilloscopes, FPGA boards, spectrum analyzers, and sensor kits—that sit idle up to **80% of the academic calendar**. Concurrently, student innovators and robotics hackathon teams face severe equipment starvation.

The IOTogether Shared Platform

IOTogether eliminates institutional silos via a unified hardware registry. Verified students borrow items under transparent duration timers; lab heads maintain total inventory control; and automated QR verification ensures physical equipment safety without administrative overhead.

2. User Personas: The Keeper & Seeker Model

02 / PROTOCOL

Keepers (Lab Custodians & Department Heads)

Primary Interface: `/sender` portal.

- **Inventory Custody:** Catalog hardware with spec sheets, manuals, daily rental/deposit rates, and training prerequisites.
- **ID Verification:** Inspect uploaded student/staff ID cards prior to approval.
- **Handover Dispatch:** Click *Manual Verified* to generate time-sensitive dynamic Handover QR tokens.
- **Incident Review:** Inspect returned equipment and log condition ratings (Pristine, Normal Wear, Damage Report).

Seekers (Student Innovators, Teachers & Makers)

Primary Interface: `/receiver` portal.

- **Cross-Campus Discovery:** Interactive multi-facet search across all institutions with category, pricing, and availability filters.
- **AI Project Recipe Engine:** Input project ideas (e.g. "IoT Weather Station") to automatically generate bill-of-materials and reserve kits.
- **Physical Handover Scan:** Scan the Lab Head's screen QR code at pickup to instantly activate loan timers.
- **Active Countdown Timers:** Live day, hour, minute, and second countdowns to return deadlines.

3. 3D Chamber & Single-Sided Cable Mechanics

03 / THREE.JS

The landing chamber features a real-time 3D simulation constructed directly with vanilla Three.js and WebGL. Rather than a double-ended cable, IOTogether employs an authentic **single-sided cable** where one end is anchored to an industrial floor terminal, and a single free grabbable plug can be physically connected to either portal socket.

Position-Based Dynamics

`RopePhysics.ts` implements a 61-node Verlet constraint solver running at 120Hz substeps. Simulates real cable sag, bend stiffness (0.045), air damping (0.994), and floor friction (0.68).

Socket 01: Keeper Portal

Located at `x: -2.05, y: 2.24` on the wall. Snapping the single plug into Socket 01 triggers the left Edison bulb into a warm golden amber glow (`#f59e0b`) and routes to `/sender`.

Socket 02: Seeker Portal

Located at `x: 1.72, y: 2.24` on the wall. Snapping the plug into Socket 02 triggers the right Edison bulb into a radiant electric cyan glow (`#06b6d4`) and routes to `/receiver`.

Single-Sided Cable Architecture Highlights

Fixed Floor Terminal: Point 0 is permanently pinned at `(0, 0.08, 2.5)` with a flanged brass ring and dark rubber boot.

Continuous Dynamic Mesh: `CableMesh.ts` rebuilds 112 rings × 8 radial facets per frame with dynamic normals for zero-artifact bending.

3D Depth Navigation: Dragging users can scroll the mouse wheel or toggle floating `+ NEAR` / `– FAR` buttons to move the plug along the Z-plane.

Procedural Audio: Web Audio API synthesizes realistic pitch-swept contact relays upon connection and subtle chirps on disconnect.

4. Physical Handover Verification Protocol

04 / LIFECYCLE

PHASE	ACTOR	ACTION & STATE MUTATION	SECURITY / AUDIT CHECKPOINT
1. Request	Seeker	Selects equipment, inputs duration (Days, Hours, Minutes), and uploads student ID card photo. State: `PENDING`	Stock concurrency check blocks overbooking atomically.
2. ID Inspection	Keeper	Reviews ID photo, verifies student enrollment, and clicks **Manual Verified**. State: `APPROVED`	Generates cryptographic time-limited Handover QR code on Lab Head screen.
3. Handover Scan	Seeker	Physically visits lab, inspects item, and scans Lab Head QR code. State: `CHECKED_OUT`	Initial condition logged. Live loan countdown timer starts ticking.
4. Return & Audit	Keeper	Performs return inspection; grades physical integrity. State: `COMPLETED`	Appends immutable entry to the cryptographic audit ledger.

5. Participating Academic Institutions

05 / SEED DATA

INSTITUTION NAME	TYPE	KEEPER PROFILE (LAB HEAD)	SEEKER PROFILE (STUDENT)
Northstar Institute of Technology	Engineering University	Dr. Aris Vance (Head of Robotics Lab)	Devin Clark (IoT Club President)
Apex Engineering College	Autonomous College	Prof. Marcus Sterling (FabLab Custodian)	Maya Patel (Robotics Lead)
Green Valley Model Senior Secondary School	Senior Secondary School	Sarah Jenkins (STEM Lab Coordinator)	Alex Chen (High School Innovator)

6. Automated Verification Test Suite

06 / VERIFICATION

Executed via `scripts/run-tests.ts` with zero failures across all critical transactional and physical lifecycle paths:

```
=== IOTOGETHER HACKATHON VERIFICATION TEST SUITE RESULTS ===
• [PASS] 1.1 Seed profiles exist and include demo roles (Keepers, Seekers, Auditors)
• [PASS] 1.2 Institutions exist with verified institutional status (2 Colleges • 1 School)
• [PASS] 1.3 Initial inventory covers free, deposit-backed, and training-restricted equipment
• [PASS] 2.1 Integer paise pricing calculations are exact without floating-point rounding drift
• [PASS] 3.1 Successful booking atomic insertion when available stock satisfies request
• [PASS] 3.2 Rejects booking when return timestamp is before or equal to start timestamp
• [PASS] 3.3 Overlapping competing bookings exceeding stock are blocked atomically
• [PASS] 4.1 Pending booking can be approved by manager
• [PASS] 4.2 Check-out records initial condition and transitions state to checked_out
• [PASS] 4.3 Return inspection transitions booking to completed and appends audit log
• [PASS] 4.4 Return with damage initiates incident review status
• [PASS] 5.1 Adding to waitlist increments queue position in strict FIFO discipline
• [PASS] 6.1 Borrower submits booking with ID photo and duration in days, hours, minutes
• [PASS] 6.2 Lab Head reviews ID photo, clicks Manual Verified, and Handover QR code generates
• [PASS] 6.3 Receiver scans Lab Head QR code, handover is confirmed, and countdown timer activates
TOTAL RESULTS: 15 PASSED • 0 FAILED • 100% SUCCESS RATE
```

7. Technology Stack & Software Design

07 / STACK

Core Framework & State

- **Framework:** Next.js 14.2.15 (App Router, Server Actions, Dynamic Routes).
- **Language:** TypeScript 5.6 with strict typing across all interfaces.
- **State Persistence:** Atomic filesystem JSON store with transaction locks.
- **Currency Arithmetic:** Stored strictly in integer paise (₹1 = 100 paise) preventing IEEE 754 floating drift.
- **Security:** Session cookies with HMAC signature & role authorization guards.

3D Rendering & UI Polish

- **Graphics Engine:** Three.js 0.170.0 (OrthographicCamera, PCF Soft Shadows).
- **Color Grading:** ACESFilmic Tone Mapping, 2048×2048 Sun Light Map.
- **Styling System:** Tailwind CSS with custom editorial fonts (Instrument Serif, DM Sans, DM Mono).
- **Dock Navigation:** Apple-style interactive spring-enlarged dock hidden on landing and active on dashboard pages.
- **Theme:** Crisp, warm ceramic studio palette (Background: `#f3efe6`, Primary: `#ba4d2b`).

8. Next.js App Router Route Manifest

08 / MANIFEST

ROUTE PATH	PURPOSE & ACCESS CONTROL	ROUTE PATH	PURPOSE & ACCESS CONTROL
`/`	3D Single-Sided Cable Interactive Chamber	`/receiver/project-recipe`	AI-Powered Hardware Recipe Builder
`/sender`	Lab Head / Equipment Custodian Dashboard	`/receiver/bookings`	Active Loans & Countdown Timers
`/sender/requests`	ID Card Inspection & Handover QR Generator	`/admin/audit`	Append-Only Cryptographic Audit Log
`/sender/inventory/new`	Equipment Registration & Spec Sheet Upload	`/notifications`	Real-Time Notification Dispatch Center

ROUTE PATH	PURPOSE & ACCESS CONTROL	ROUTE PATH	PURPOSE & ACCESS CONTROL
/receiver/explore	Cross-Institutional Search & Filter Catalog	/profile	Academic Membership & ID Card Profile